

Antennas

At the end of this episode, I will be able to:

1. Antennas

Learner Objective: *Learn about some of the various antennas used in wireless networking today*

Description: In this episode, you will learn about just some of the wireless antennas used in modern wireless networks today. These include omnidirectional versus directional antennas.

-
- Antennas play a crucial role in wireless networking by transmitting and receiving radio frequency signals.
 - The choice of antenna type depends on the specific requirements of the wireless network, such as coverage area, signal range, and directionality. Here are some common antenna types found in wireless networking today:
 - Omnidirectional Antennas - radiate signals in all directions, creating a 360-degree coverage pattern. They are well-suited for providing coverage in all directions from a central point.
 - Directional Antennas - focus the signal in one specific direction, providing a more concentrated and longer-range coverage in that direction. They are used to establish point-to-point or point-to-multipoint links.
 - Yagi Antennas - Yagi antennas are a type of directional antenna with multiple elements arranged in a line. They are often used for point-to-point communication over moderate distances.
 - Parabolic Dish Antennas - use a dish-shaped reflector to focus radio waves onto a central point. They are highly directional and commonly used for long-distance point-to-point links.
 - Patch Antennas - also known as panel antennas, have a flat, rectangular shape. They are often used in indoor environments and can be directional or omnidirectional depending on their design.
 - Sector Antennas - provide coverage in a specific sector or wedge-shaped pattern. They are commonly used to cover larger outdoor areas with a directional focus.
-

Additional Resources:

Antenna

[https://en.wikipedia.org/wiki/Antenna_\(radio\)](https://en.wikipedia.org/wiki/Antenna_(radio))